

# Modeling the effects of thermal and mechanical load cycling on a C/SiC composite in oxygen/argon mixtures

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## Abstract

Analytical solutions of and experimental results on the strain response of a carbon fiber reinforced SiC matrix composite under thermal and mechanical load cycling in O<sub>2</sub>/Ar are presented. Thermal strain and mechanical strain were shown to approximately sustain linear relationships with temperature  $T$  and stress  $\sigma$ , respectively; whereas baseline strain was considered to be damage-dependent, resulting from a combination of two major contributing mechanisms: (a) a physical mechanism in the form of matrix microcracking accompanied by fiber debonding, sliding or fracture and (b) a chemical mechanism in the form of the fiber oxidation associated with longitudinally increased compliance. Based on these analyses a theoretical model, taking into account the thermal strain, mechanical strain and baseline strain, was theoretically formulated with respect to the contribution of each on the overall total strain and to their generation mechanisms. The proposed model gave correct and reliable predictions.

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## 1. Introduction and overview

The application of ceramic matrix composites (CMCs) to advanced airframe and propulsion systems for future space transportation vehicles can provide benefits in life, performance, temperature margin, and weight savings [1–3]. One CMC system of interest to the aerospace community is carbon fiber reinforced silicon carbide (C/SiC). Compared to monolithic ceramics, this material presents higher toughness and tolerance to the presence of cracks, which implies a non-catastrophic mode of failure. But one of the more formidable obstacles to the widespread use of C/SiC structures is that the carbon fibers within the cracked matrix oxidize at medium to high temperatures in oxidizing environments, especially in presence of external temperature impacts and/or mechanical stress variations [4–6]. It is therefore necessary to develop a tool that is capable of determining the extent of oxidation and the residual strength and stiffness in the C/SiC component as a function of the time, temperature, stress and environmen-

tal oxygen concentrations to which the C/SiC structure is exposed.

In this regard, much work has been done to describe and to model the oxidation of fibres, matrices and interfaces in the absence of loading by assuming steady state diffusion of the oxidant to the site of oxidation [7–9]. Additionally, several stress-oxidation models have also been developed to investigate the effects of the oxidation of constituents on the mechanical response or life of the composite material accounting for the synergistic creep-oxidation interactions at high temperature [6,10–13]. Although these previous studies have provided insight into the physics and mathematics of carbon phase oxidation in ceramic composites under both unstressed and stressed conditions, these approaches are not readily and directly applicable to support the complicated strain response analysis of C/SiC composites subjected to thermal and mechanical load cycling in oxidizing environments. Thus, a new model that treats the influences of thermal, mechanical and chemical applied conditions on the strain evolution of C/SiC composite is needed.

Recently, Mei et al. [14–16] have experimentally reported some novel findings concerning real-time strain

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response of a C/SiC composite which was subjected simultaneously to thermal cycling and mechanical stress in oxidizing atmosphere using an induction heating environmental chamber fixed on the Instron servo-hydraulic tester. Many of these investigations have implied that increase in damage strain of the C/SiC composite was ascribed to the result of the combination of the following two major contributing mechanisms:

- Physical damage mechanism, thermal cycling and mechanical stress induced matrix microcracking normal to tensile axis, known as crack opening displacement, accompanied by interfacial debonding or sliding (see Figs. 1 and 2 in SM<sup>1</sup>).
- Chemical damage mechanism, longitudinally increased compliance owing to reduction in the effective load bearing area resulting from the progressive oxidation of the reinforcing fibers from surface to interior through the cracks created earlier (see Fig. 3 in SM).

The aim of the present work is to incorporate the effects of the thermophysical and chemical environmental variables into a formulated strain response model on basis of the above damage mechanisms. In this paper, a strain evolution model for C/SiC composites subjected to thermal and mechanical load cycling in an O<sub>2</sub>/Ar mixture was presented. The model mainly accounted for the crack propagation and multiplication of the brittle SiC matrix, and the oxidation of the reinforcing C fibers, both of which increased the baseline strain of the material with time. At the same time, thermal strain owing to temperature variations and mechanical strain owing to stress changes were also taken into account and formulated. The proposed model, without any attempt at fitting, gave correct and reliable predictions in assessing the experimentally obtained strain response behaviour for a specific case of a 3D C/SiC composite subjected to repetitive temperature between 900 and 1200 °C and cyclic load of 60 ± 20 MPa in a 10.4 vol.% O<sub>2</sub>/89.6 vol.% Ar mixture.

## 2. Experimental description

### 2.1. Materials

As described in [14], the same isothermal CVI process was employed to fabricate the 3D C/SiC composites at ~1000 °C in this investigation. Fiber architectures in the as-fabricated C/SiC composite preforms were shown in Fig. 4 in SM. The dog-bone shaped specimens with a dimension of 185 × 3 × 3 mm<sup>3</sup> were cut from the fabricated composite plates, and further coated with SiC by the same I-CVI processing (thickness ≈ 50 μm). Table 1 in SM summarized the major properties of the as-received specimens. Fig. 5 in SM shown a typical tensile stress–strain curve of the 3D C/SiC composite specimens. It can be seen that the composite behaves as a typical damageable material, exhibiting an extensive non-linear

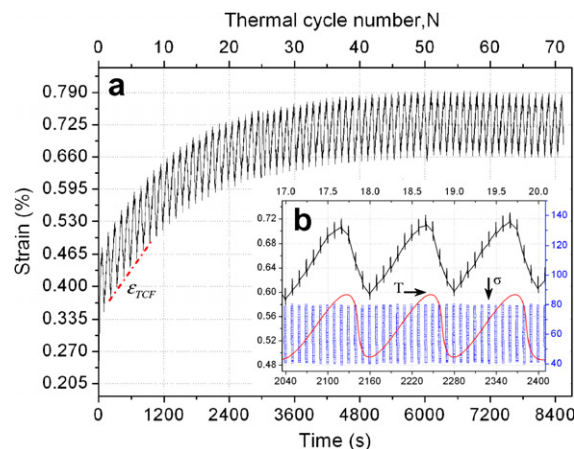


Fig. 1. (a) The measured strain curve of the C/SiC composite subjected simultaneously to thermal cycling and fatigue stress in O<sub>2</sub>/Ar mixture and (b) a close-up of strain within several cycles. The red broken line represents the baseline strain. (For interpretation of the references in colour in this figure legend, the reader is referred to the web version of this article.)

stress–strain domain up to rupture. The linear deformation is limited up to about 50 MPa (referred to as first-matrix cracking stress  $\sigma_c$ ), after which the behaviour becomes non-linear.

### 2.2. Thermal cycling tests with external stress

Thermal cycling experiments were conducted in an oxygen/argon mixture of 10.4 vol.% O<sub>2</sub>/89.6 vol.% Ar using a newly developed integrated system, which was described in detail in Fig. 2 of [14]. Thermal cycling was carried out between temperatures of  $T_{\text{lower}} \approx 900$  °C and  $T_{\text{upper}} \approx 1200$  °C ( $\Delta T \approx 300$  °C) over a period of 120 s and the mean heating/cooling rates were about 5 K/s and –10 K/s, respectively. During the testing, a small fatigue stress of 60 ± 20 MPa<sup>2</sup> (sinusoidal wave, frequency = 1 Hz, stress ratio  $R = 0.5$ ) was simultaneously applied to the dog-bone specimens. Strain was measured directly from the gauge length by a contact Instron extensometer (Model A1452-1001B). Microstructures of the specimens were observed with a scanning electron microscope (SEM, Hitachi S-4700).

## 3. Strain results and analysis

Following the test procedure described above, a typical strain response curve of the tested C/SiC composite specimen to cyclic temperature and fatigue stress was obtained and plotted in Fig. 1. Fig. 1b is a close-up observation of the strain curve within several cycles from the No. 17th to 20th cycle (i.e., from 2040 s to 2400 s). It should be noted from Fig. 1 that the total strain should be a combined result of thermal strain due to heating/cooling, mechanical strain due to loading/unloading and baseline strain due to damage accumulation. As a representative, the strain response data within the No. 18th thermal cycle is summarized in Table 1. It can be seen from Fig. 1b and Table 1, thermal strain and mechanical strain linearly increase with

<sup>1</sup> Some figures and data from the previous investigations were put in the Supplementary material (SM) section.

<sup>2</sup> A relatively small stress is usually selected to apply on the C/SiC composite because the reinforcing fibers is susceptible to oxidation once the cracks open.

Table 1

Strain response results of the C/SiC composites to thermal cycle and fatigue stress within the No. 18th thermal cycle

| No.  | $T$                 | $\dot{T}(t)$ (K/s) | $\dot{\epsilon}_{Th}(t)$ ( $s^{-1}$ ) | $\epsilon_{Th}$ (%) | $\epsilon_{Th}^R$ (%) | $\sigma$ (MPa) | $\epsilon_{Fat}^{max}-\epsilon_{Fat}^{min}$ (%) | $\epsilon_{Fat}^R$ (%) |
|------|---------------------|--------------------|---------------------------------------|---------------------|-----------------------|----------------|-------------------------------------------------|------------------------|
| 18th | Heating 900–1200 °C | 5                  | $1.95 \times 10^{-5}$                 | 0.594–0.711         | 0.117                 | $60 \pm 20$    | 0.682–0.658                                     | 0.024                  |
|      | Cooling 1200–900 °C | –10                | $-3.67 \times 10^{-5}$                | 0.711–0.601         | 0.110                 | $60 \pm 20$    | 0.663–0.634                                     | 0.029                  |

$\epsilon_{Fat}^{max}-\epsilon_{Fat}^{min}$  gives the mean value of 12 maximum–minimum fatigue strains within the No. 18th thermal cycle.

increase of  $T$  or  $\sigma$  and then linearly decrease with decrease of  $T$  or  $\sigma$ . As cycle proceeds, the thermal strain and mechanical strain are periodically repeated with the fixed strain ranges and as the same period as their excitation temperature or stress, although the baseline strain is continuously increasing. These experimentally observed results are advantageous in modeling the effects of temperature and stress on the strain.

### 3.1. Thermal strain

Assuming that thermal strain is absolutely temperature-dependent and follows a linear function as the temperature  $T$

$$\epsilon_{Th}(t) = \alpha T(t) \quad (1)$$

and, its rate can be expressed as

$$\dot{\epsilon}_{Th}(t) = \frac{\partial \epsilon_{Th}(t)}{\partial t} = \alpha \frac{\partial T(t)}{\partial t} = \alpha \dot{T}(t) \quad (2)$$

where  $\dot{T}(t)$  is the heating/cooling rate,  $\alpha$  signifies the coefficient of thermal expansion (CTE) along the composite material axis direction. If the composite material is thermally cycled between two selected temperatures, the thermal strain range  $\epsilon_{Th}^R$  between the lower and upper temperatures at each cycle is determined as,

$$\epsilon_{Th}^R = \epsilon_{Th}^{T_{upper}} - \epsilon_{Th}^{T_{lower}} = \bar{\alpha} \Delta T \quad (3)$$

$\bar{\alpha}$  is the mean CTE between two temperatures, and  $\Delta T$ , temperature difference.

### 3.2. Mechanical strain

Generally, a relatively small stress can be considered to apply on the C/SiC composite because it is well-known that the reinforcing carbon fibers is susceptible to oxidation once the cracks open. When an applied stress is below the so-called proportional limit (i.e.,  $\sigma_c$ ), the mechanical strain of the composite is linear elastic and dependent on the stress as the classical Hooke's law

$$\epsilon_{Me} = \frac{\sigma}{E} \quad (4)$$

Under a cyclic stress, the mechanical strain range can be obtained through

$$\epsilon_{Me}^R = \frac{\Delta \sigma}{E} \quad (5)$$

where,  $E$  is the Young's modulus of the composite and  $\Delta \sigma$ , stress range.

### 3.3. Baseline strain

The baseline strain, in fact, is a retained strain of the composite specimen once unloading both thermal and mechanical loads to the bottoms of temperature and of stress. In this paper, it can be specially defined as thermal cycling fatigue strain (hereafter, TCF strain)

$$\epsilon_{TCF} = \epsilon_{T_{bottom}}^{\sigma_{bottom}} \quad (6)$$

and its rate can be written as

$$\dot{\epsilon}_{TCF}(t) = \frac{\partial \epsilon_{TCF}(t)}{\partial t} \quad (7)$$

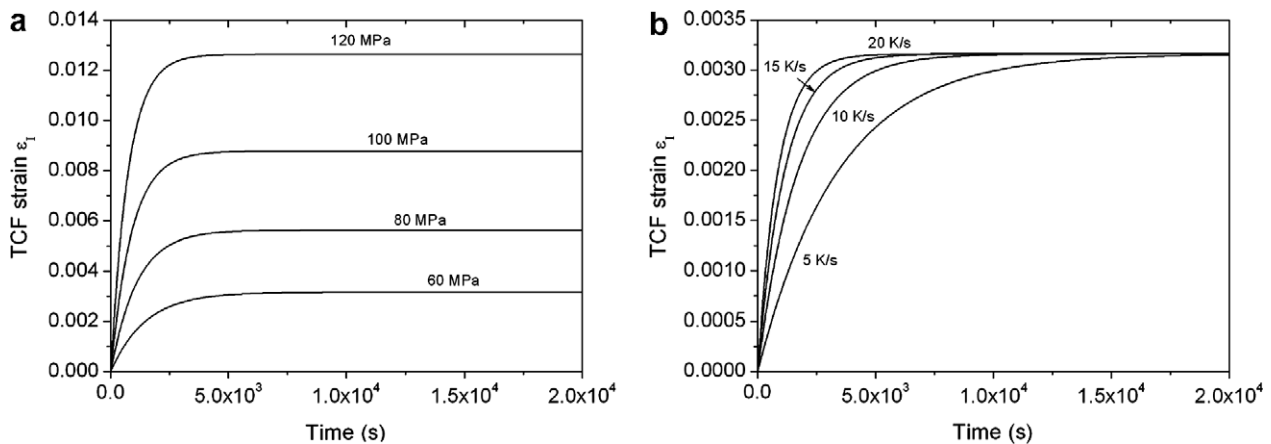


Fig. 2. The I TCF strain as a function of test time where the strain increase is governed by a physical damage mechanism, (a) at different stress of 60, 80, 100, 120 MPa and (b) in different cooling rate of 5, 10, 15, 20 K/s.

Hence, the TCF strain can be acceptable to act as a damage indicator when the CMC composites are subjected to thermal cycling and mechanical stress in an oxidizing atmosphere. As schemed in the Fig. 1, the slope of the red broken line represents the mean rate of the TCF strain. The higher the rate, the faster the TCF strain increases and the severer the damage to the composites. If the rate becomes zero with continued cycles, the composites go to a steady state in which initiation of the new cracks or propagation of the previous cracks is terminated transitorily. A progressively increasing TCF strain rate means that the composite will fracture eventually.

#### 4. Model formulation

The derivation of the model relies on the principle that the total strain response behaviour of the composite can be expressed as the sum of three terms: i.e., thermal strain, mechanical strain and TCF strain. For the formulation of the model, the former two terms are simplified to follow the linear laws as Eqs. (1) and (4). Thus, the development of the model will focus on the better understanding the physical and chemical mechanisms of the successive TCF strain evolution during thermal cycling and mechanical stress in oxidizing atmosphere.

##### 4.1. Matrix cracking and fiber sliding contribution

Consider a model composite of length  $L$ , that is reinforced with fibers of equal diameter  $d$  that occupy a fraction  $V_f$  of the composite volume. It is assumed that the fibers are aligned along the axis at braiding angles  $\phi$ , and coated with a proper PyC layer of finite thickness to impart composite behaviour to the CFCC and that the distribution of tensile strengths for the fibers and the fiber interfacial shear stress,  $\tau$ , are uniform and known. The matrix is assumed to no plastic deformation and the inelastic strain is mainly derived from the sum of crack opening displacement of transverse crack system, accompanied by the fiber debonding or sliding along a finite region adjacent to the matrix crack. It is also assumed that the composite consists of a matrix of volume fraction  $V_m$  with properties  $E_m$  (Young's modulus),  $\alpha_m$ , which contains the reinforcing fibers with properties  $E_f$ ,  $\alpha_f$ .

Consider the case when such the composite is subjected to thermal cycling and a tensile stress, larger than the matrix cracking stress,  $\sigma_c$ , so that a series of parallel and equally spaced transverse cracks are formed in the matrix (see Fig. 1 in SM). The TCF strain is simply the density of transverse matrix microcracks  $\beta$  (i.e., number of cracks per m) multiplied by the crack opening displacement of each crack  $\delta$ , as

$$\varepsilon_{\text{TCF}}^{\text{I}} = \frac{\Delta L^{\text{in}}}{L} = \frac{n \cdot U_{\text{COD}}}{L} = \delta \cdot \beta \quad (8)$$

where  $n$  is the number of transverse cracks and  $U_{\text{COD}}$ , crack opening displacement. Crack opening displacement of each crack  $\delta$  can be simply estimated as [17]

$$\delta = \frac{dV_m^2 E_m^2 \cos \phi}{4V_f^2 \tau E_f (E_f V_f + E_m V_m)^2} \sigma_A^2 \quad (9)$$

where  $\sigma_A$  is the applied stress (MPa) and  $\tau$ , the shear sliding stress of interface (MPa). The crack growth and multiplication are taken into account to follow the exponential law as

$$\beta = \eta \beta_s (\alpha_m - \alpha_f) \Delta T \left\{ 1 - \exp \left( \left[ \frac{\lambda \sigma_A E_m V_m (\alpha_f - \alpha_m)}{(E_m V_m + E_f V_f)^2} \dot{T} \right] t \right) \right\} \quad (10)$$

where  $\beta_s$  is the saturated crack density ( $\text{m}^{-1}$ ),  $\eta$  and  $\lambda$  are the constants,  $\dot{T}$  is the maximum heating/cooling rate (K/s). Substituting Eqs. (9) and (10) into (8) gives the following formulation of the type-I TCF strain for the physical damage mechanism,

$$\varepsilon_{\text{TCF}}^{\text{I}} = \frac{\sigma_A^2 V_m^2 E_m^2 d \eta \beta_s (\alpha_m - \alpha_f) \Delta T \cos \phi}{4V_f^2 \tau E_f (E_f V_f + E_m V_m)^2} \times \left\{ 1 - \exp \left( \left[ \frac{\lambda \sigma_A E_m V_m (\alpha_f - \alpha_m)}{(E_m V_m + E_f V_f)^2} \dot{T} \right] t \right) \right\} \quad (11)$$

Note that the above type-I TCF strain is a time-dependent function where the strain magnitude and shape are governed by the environmental parameters, i.e.,  $\sigma_A$ ,  $\dot{T}$ ,  $\Delta T$ ,

Table 2  
Parameters and values used in calculation

| Parameter                                 | Symbol         | Value    | Units                   |
|-------------------------------------------|----------------|----------|-------------------------|
| <i>Material</i>                           |                |          |                         |
| Young's modulus of matrix                 | $E_m$          | 350      | GPa                     |
| Volume fraction of matrix                 | $V_m$          | 0.6      |                         |
| CTE of matrix                             | $\alpha_m$     | 4.6      | $10^{-6}/\text{K}$      |
| Young's modulus of fiber                  | $E_f$          | 230      | GPa                     |
| Volume fraction of fiber                  | $V_f$          | 0.4      |                         |
| CTE of fiber                              | $\alpha_f$     | 0.5      | $10^{-6}/\text{K}$      |
| Fiber diameter                            | $d$            | 7        | $\mu\text{m}$           |
| Side of the square specimen cross section | $a$            | 0.003    | m                       |
| Braiding angle                            | $\phi$         | 22       | $^\circ$                |
| <i>Environmental</i>                      |                |          |                         |
| Applied stress                            | $\sigma_A$     | 60       | MPa                     |
| Fatigue stress range                      | $\Delta\sigma$ | 40       | MPa                     |
| Constant                                  | $\eta$         | 14,346   |                         |
| Constant                                  | $\lambda$      | 120,050  |                         |
| Oxidant partial pressure                  | $\chi$         | 0.104    | %                       |
| Total pressure                            | $P$            | 101,325  | Pa                      |
| Molar density of carbon                   | $\rho_c$       | 150,000  | $\text{mol}/\text{m}^3$ |
| Gas constant                              | $R$            | 8.31441  | $\text{J}/\text{mol K}$ |
| Lower temperature limit $T_0$             | $T$            | 1173     | K                       |
| Heating/cooling rate                      | $\dot{T}$      | 5/–10    | K/s                     |
| Thermal cycling temperature difference    | $\Delta T$     | 300      | K                       |
| <i>Micromechanical</i>                    |                |          |                         |
| Saturated crack density                   | $\beta_s$      | 7000     | $\text{m}^{-1}$         |
| Crack opening displacement                | $\delta$       | Variable | m                       |
| Sliding resistance of interface           | $\tau$         | 3        | MPa                     |

CTE, Coefficient of thermal expansion.

etc., associated with the physical damage mechanism by assuming that the material parameters remain constant.

Due to page limitation, the effect laws of only those representative environmental parameters on the strain are calculated and plotted in the following. The readers may predict other controlled laws of the environmental parameters of interest by using the proposed models. Using the data listed in Table 2, TCF strain predictions were generated for a specific case of the C/SiC composite. Fig. 2 shows the predicted type-I TCF strains for tests conducted at different applied stresses of 60, 80, 100, 120 MPa (Fig. 2a) and in different cooling rate of 5, 10, 15, 20 K/s (Fig. 2b). The effect of stress can be seen clearly in Fig. 2a. As expected, the higher the applied stress, the shorter the time to crack saturation and the larger the TCF strain to saturation (defined as a specific inflexion strain whose first derivative is equal to zero). As stress increases linearly, the TCF strain to saturation is promoted quadratically. It can be also seen from Fig. 2b that the higher the cooling rate, the greater the TCF strain rate and the shorter the time to crack saturation. In summary, the parameters in the amplitude of the Eq. (11), e.g. the applied stress, significantly influence the magnitude of the TCF strain while the parameters in the exponential argument, e.g. the heating/cooling rate, mainly change its shape.

#### 4.2. Fiber oxidation contribution

Matrix cracks will serve as avenues for the ingress of the environment atmosphere into the composite. In this analysis we are concerned with the effects of oxidizing environments on the reliability of a CMC through reacting with the carbonaceous fibers in microcracks and the oxidation of the matrix is neglected.

When the oxidizing gas ingresses into the composite a sequence of events is triggered starting first with the oxidation of the fiber. Especially, in presence of a small applied stress  $\sigma_A$ , the fiber oxidation can result in increase of the TCF strain by reducing the effective load bearing area of the reinforcing fibers in a square cross-section of the composite specimen with a side length of  $a$  (as schemed in Fig. 3). Therefore, the type-II TCF strain will increase with the reduction both in effective cross-section area  $A(t)$  and in the volume-averaged Young modulus by

$$\begin{aligned}\varepsilon_{\text{TCF}}^{\text{II}} &= \frac{\sigma_A A_0}{\left[ \frac{A(t)}{A_0} (E_f V_f + E_m V_m) \right] A(t)} \\ &= \frac{\sigma_A a^4}{(E_f V_f + E_m V_m) [a - 2x/(V_f \cos \varphi)]^4}\end{aligned}\quad (12)$$

Equivalently,

$$\varepsilon_{\text{TCF}}^{\text{II}} = \frac{\sigma_A}{(E_f V_f + E_m V_m) [1 - 2x/(a V_f \cos \varphi)]^4} \quad (13)$$

where  $x$  is the recession distance of carbon fibers from the surface into the interior of the composite (as also schemed in Fig. 3). In the present study, the oxidation of carbon phase is assumed to be controlled by diffusion of oxygen gas through the matrix microcracks. In this case, Eckel et al. [7] suggested that the recession distance  $x$  can be properly developed, related to the exposure time  $t$ , as

$$x^2 = K_p t = \left( 4D \frac{P}{\rho_c R T} \ln \frac{(1 + \chi)(D_k/D) + 1}{D_k/D + 1} \right) t \quad (14)$$

where  $K_p$  is referred to as the oxidation rate.  $\chi$  is the oxidant partial pressure,  $P$  is the total pressure (Pa),  $\rho_c$  is the molar density of carbon ( $\text{mol/m}^3$ ),  $R$  is the gas constant ( $\text{J/mol K}$ ),  $T$  is the absolute temperature (K),  $D_k$  and  $D$  are Knudsen diffusion coefficient and Fick diffusion coefficient, respectively. The type-II TCF strain, integrating Eqs. (13) and (14), can be rewritten as

$$\varepsilon_{\text{TCF}}^{\text{II}} = \frac{\sigma_A}{(E_f V_f + E_m V_m) [1 - 2\sqrt{K_p t}/(a V_f \cos \varphi)]^4} \quad (15)$$

Eckel et al. also suggested that the  $K_p$  may be simply developed as a function of the oxidant partial pressure  $\chi$ , the temperature  $T$  and the characteristic dimension of the crack opening displacement  $\delta$

$$K_p = 6.263 \times 10^{-10} T^{1/2} \ln \left[ \frac{3.8 \times 10^6 (1 + \chi) \delta + 1}{3.8 \times 10^6 \delta + 1} \right] \quad (16)$$

As mentioned previously,  $\delta$  could be mainly determined by the external applied stress  $\sigma_A$  according to Eq. (9). The increased stresses widen the crack and the increased temperatures narrow the cracks. Using the Eq. (16), the effects of dimension of  $\delta$  on oxidation rate  $K_p$  at different temperature and in different oxidant partial pressure are plotted in Fig. 4a. It is evident that temperature only weakly affects the magnitude and shape of the  $K_p$  curve in the small  $\delta$  ( $\leq \delta_c = 1 \mu\text{m}$ ) regime. In the large  $\delta$  ( $\geq \delta_c$ ) regime, the speed of gas diffusion is much greater than the reaction rate of the

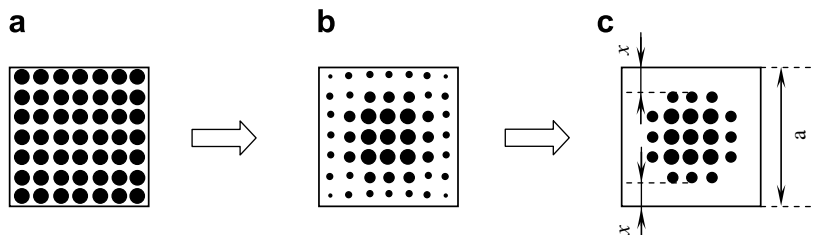


Fig. 3. Schematic diagrams of fiber oxidation in the cross-section of C/SiC composites.

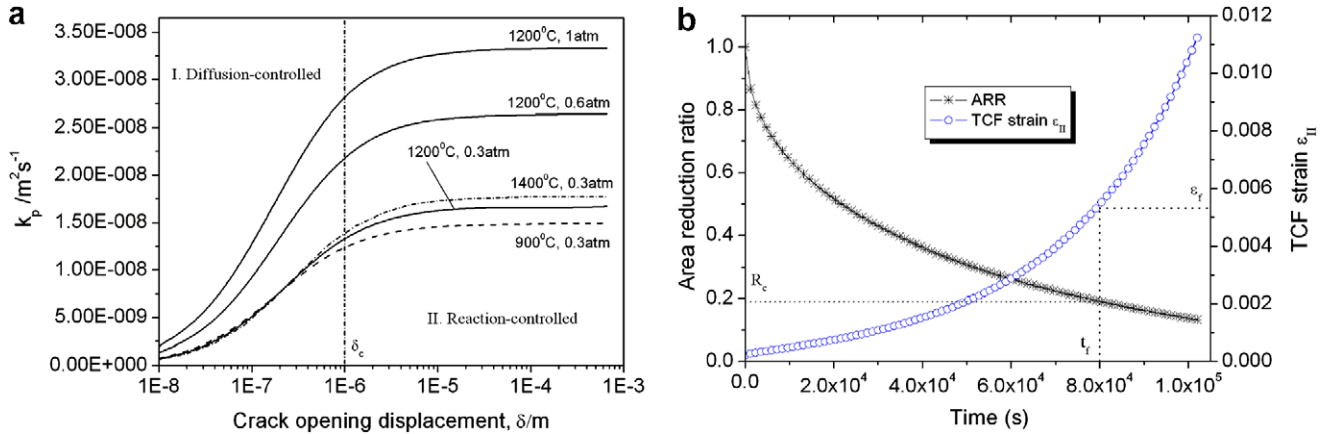


Fig. 4. (a) Effect of crack opening displacement  $\delta$  on recession rate  $K_p$  and (b) relationships of the II TCF stain and ARR of the composite with test time  $t$  under the stress of 80 MPa and the oxidant partial pressure of 0.1%.

carbon phase leading to the transformation from the diffusion-controlled to the reaction-controlled regime. Increased oxidant partial pressure is shown to accelerate the gas diffusion both in small and large  $\delta$  regimes. Thus,  $\delta$  becomes a significant microstructural parameter and plays a key role in the oxidation kinetics mechanism of carbon phase (hereafter, “ $\delta$ -controlled regime”). Environmental temperature and stress change the oxidation kinetics mechanism of the carbonaceous CMCs by firstly changing the characteristic dimension of the  $\delta$ .

Fig. 4b illustrates relationships of the predicted type-II TCF strain using the above model of Eq. (15) and the effective load bearing area reduction ratio (ARR, i.e.  $A(t)/A(0)$ ) with test time  $t$  under the stress of 80 MPa and the oxidant partial pressure of 0.1%. It is obvious that there exists a close correlation between the ARR and the type-II TCF strain, i.e., where the type-II TCF strain increases most, the ARR of the composite decreases most as well. This is a strong indication that the reduction in the effective load bearing area of the fibers plays a critical role in the increased compliance behaviour of the C/SiC composites exposed in the oxidizing atmosphere. This process can be best described using the ‘snow ball effect’ analogy. When the fibers are subjected to a chemical recession at a constant tensile stress, the superficial carbon fibers will fail firstly, and then because of global load sharing assumptions, the load originally carried by the now broken fibers will be transferred to the surviving fibers. Consequently, the surviving fibers are now subjected to a larger tensile stress. The composite will eventually fail when the progressively increasing tensile stress reaches the UTS of the materials with decreasing ARR as

$$R_c = \frac{A(t)}{A_0} = \frac{\sigma_{Amax}}{\sigma_{UTS}} \quad (17)$$

where  $R_c$  signifies the critical ARR,  $\sigma_{Amax}$  represents the maximum value of the applied stress. That is to say, in the present experiment the tested C/SiC composite will fail once  $R_c$  reaches 0.193 ( $\sigma_{Amax} = 80$  MPa and  $\sigma_{UTS} =$

413.76 MPa), at which point the time to failure  $t_f \approx 22.2$  h (i.e., 80,000 s) and the type-II TCF strain to failure approximates to 0.53% as determined in Fig. 4b. Through the model of Eq. (15) and the data listed in the Table 2, predicted distributions of the type-II TCF strain at different stress of 40, 60, 80, 100, 120 MPa (Fig. 5a) and in different oxidant partial pressure of 0.1, 0.3, 0.5, 0.7 (Fig. 5b) are obtained. The effect laws of the various stresses on the type-II TCF strain in Fig. 5a are qualitatively similar to the oxidation-assisted stress-rupture results of the C/SiC composites obtained experimentally by Halbig et al. [6]. Namely, the higher the applied stress, the shorter the time to failure and the faster the TCF strain increases. Note that these curves, similar to the classical creep of the stressed metals, also show three well-defined regimes: transient, steady-state, and tertiary stages. During the transient stage the initial sudden loading is responsible for the linear increase in TCF strain. Hence, these TCF strain curves start from the different primary strain by the different stress levels separating them. The steady-state stage exhibited by the TCF strain curves indicates that the composite experiences less change in compliance and its duration is as long as the time required to slowly consume carbon fibers through the cyclic opening-closing matrix microcracks. The tertiary stage is associated with an accelerated deformation as a result of the oxidation-assisted failure of a large number of fibers. Failure of the composite occurs when a critical number of fibers fail leading to the critical ARR of  $R_c$ .

It is evident from Fig. 5b that the higher oxidant partial pressure can directly promote the TCF strain rate and shorten the time to failure by solely affecting the parabolic rate constant  $K_p$  according to Eq. (15). It can be actually observed that the same failure strain of 0.53% determined by the critical ARR corresponds to the different time to failure of  $t_{f1} \approx 22.2$ ,  $t_{f2} \approx 7.6$ ,  $t_{f3} \approx 4.6$ ,  $t_{f4} \approx 3.3$  h in the horizontal abscissa for the different oxidant partial pressure of 0.1, 0.3, 0.5 and 0.7. Additionally, as oxidant partial pressure increases its effect on the time to failure is remarkably weaken, taking on an apparent deactivation because

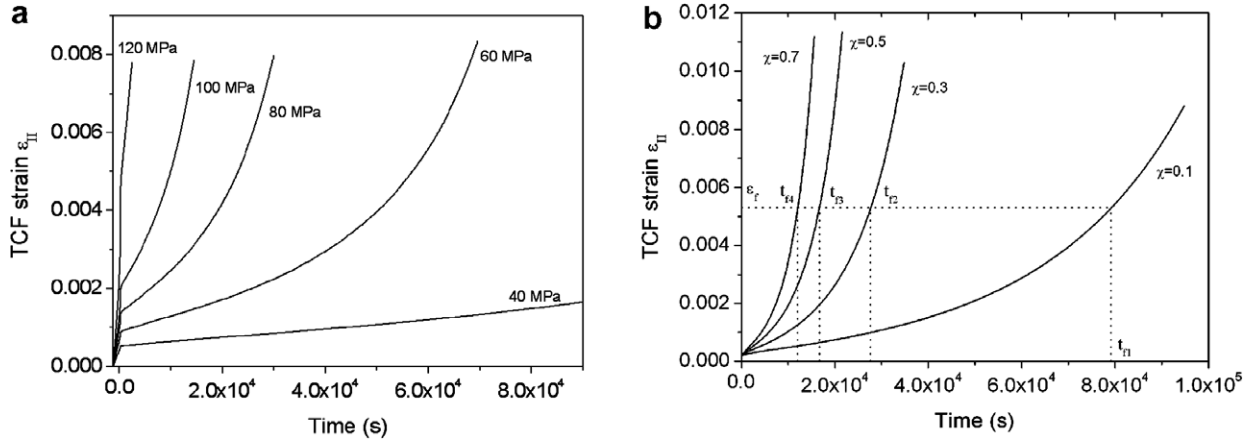


Fig. 5. The type-II TCF strain as a function of test time where the strain increase is governed by a chemical damage mechanism; (a) at different stress of 40, 60, 80, 100, 120 MPa and (b) in different oxidant partial pressure of 0.1, 0.3, 0.5, 0.7.

the effect of the oxidizing gas concentration is restrained in the diffusion-controlled kinetics regime.

#### 4.3. Total strain response model

The theoretically expected TCF strain response behaviour of the composite can be expressed as the sum of Eqs. (11) and (15)

$$\begin{aligned} \varepsilon_{\text{TCF}}^T = & \frac{\sigma_A^2 V_m^2 E_m^2 d\eta\beta_s(\alpha_m - \alpha_f)\Delta T \cos \varphi}{4V_f^2 \tau E_f (E_f V_f + E_m V_m)^2} \\ & \times \left\{ 1 - \exp \left( \left[ \frac{\lambda \sigma_A E_m V_m (\alpha_f - \alpha_m)}{(E_m V_m + E_f V_f)^2} \dot{T} \right] t \right) \right\} \\ & + \frac{\sigma_A}{(E_f V_f + E_m V_m) [1 - 2\sqrt{K_p t} / (a V_f \cos \varphi)]^4} \end{aligned} \quad (18)$$

However, when  $t = 0$ , the total TCF strain gives

$$\varepsilon_{\text{TCF}}^T|_{t=0} = \frac{\sigma_A}{(E_f V_f + E_m V_m)} \quad (19)$$

It must be notable that the thermal expansion of the composite because of the initial heating up to the lower temperature limit  $T_0$  in the first thermal cycle is not taken into account in Eq. (18). Consequently, considering the effect of the initial thermal strain  $\varepsilon_{\text{Initial}}$  yields

$$\begin{aligned} \varepsilon_{\text{TCF}}^T = & \varepsilon_{\text{TCF}}^I + \varepsilon_{\text{TCF}}^{\text{II}} + \varepsilon_{\text{Initial}} \\ = & \frac{\sigma_A^2 V_m^2 E_m^2 d\eta\beta_s(\alpha_m - \alpha_f)\Delta T \cos \varphi}{4V_f^2 \tau E_f (E_f V_f + E_m V_m)^2} \\ & \times \left\{ 1 - \exp \left( \left[ \frac{\lambda \sigma_A E_m V_m (\alpha_f - \alpha_m)}{(E_m V_m + E_f V_f)^2} \dot{T} \right] t \right) \right\} \\ & + \frac{\sigma_A}{(E_f V_f + E_m V_m) [1 - 2\sqrt{K_p t} / (a V_f \cos \varphi)]^4} \\ & + \frac{E_m V_m \alpha_m + E_f V_f \alpha_f}{E_m V_m + E_f V_f} T_0 \quad \text{and} \\ K_p = & 6.263 \times 10^{-10} T^{1/2} \ln \left[ \frac{3.8 \times 10^6 (1 + \chi) \delta + 1}{3.8 \times 10^6 \delta + 1} \right] \end{aligned} \quad (20)$$

As demonstrated through Eq. (20), the total modelled TCF strain behaviour of the composite depends on

- (1) Environmental parameters including space such as  $\sigma_A$ ,  $\dot{T}$ ,  $T_0$ ,  $\Delta T$  and  $\chi$ , etc.), and time  $t$ .
- (2) Material parameters, i.e., the properties of the matrix such as  $E_m$ ,  $\alpha_m$ , of the fibres such as  $E_f$ ,  $\alpha_f$ , braiding angle  $\phi$  and fiber diameter  $d$ , and of the composite such as volume fractions,  $V_m$  and  $V_f$ .
- (3) Micromechanical parameters associated with the interfaces and cracks, such as the interfacial shear stress  $\tau$ , crack opening displacement  $\delta$  and the saturated crack density,  $\beta_s$ .

The total TCF strain distribution at the different stress of 60, 80, 100, 120 MPa and its constitutive principle, taking from an example at 100 MPa, are illustrated in Fig. 6. As shown in Fig. 6a, it is easy to understand the contributions of the initial heating strain  $\varepsilon_{\text{Initial}}$ , the physical damage strain  $\varepsilon_{\text{TCF}}^I$  and the chemical damage strain  $\varepsilon_{\text{TCF}}^{\text{II}}$  to the total TCF strain  $\varepsilon_{\text{TCF}}^T$ . The similar three stages can be introduced to interpret the total TCF strain evolution with respect to the controlling laws of each mechanism.

Firstly, the sudden strain increase of the composites in the first cycle is ascribed in large part to the initial thermal expansion strain up to the selected lower temperature limit. Of course, another slight reason for this phenomenon should be the first loading which is taken into account in the chemical damage strain  $\varepsilon_{\text{TCF}}^{\text{II}}$  when  $t = 0$ . Secondly, thermal cycling induced matrix cracking, multiplication, fiber debonding and sliding can be considered to be responsible for the secondary larger exponential-like strain growth until the physical damage reaches saturated. Finally, the tertiary stage is associated with a continuous and slow increase in the compliance of the composite as a result of oxidation-assisted fiber failure in a  $\delta$ -controlled kinetics regime. Evidently, the three sequential stages are governed by the initially selected lower temperature limit  $T_0$  and applied stress level  $\sigma_0$ , by the subsequently thermal and mechanical cycling induced physical damage, and by the

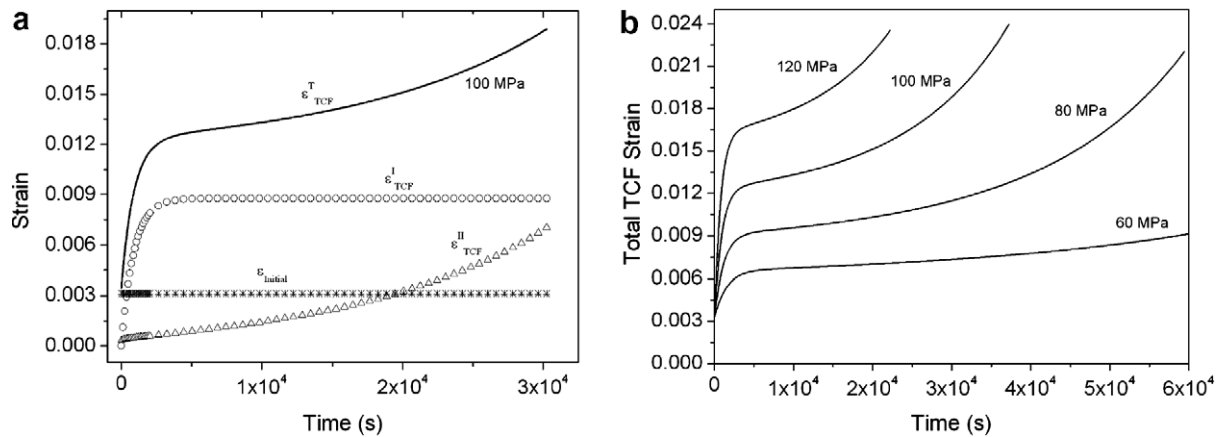


Fig. 6. (a) Contributions of the initial heating strain  $\epsilon_{\text{initial}}$ , physical damage strain  $\epsilon_{\text{TCF}}^I$  and chemical damage strain  $\epsilon_{\text{TCF}}^{II}$  to the total TCF strain  $\epsilon_{\text{TCF}}^T$ . (b) A family of the total TCF strain vs. time curves using the comprehensive TCF strain model at the different stress of 60, 80, 100, 120 MPa.

continuously oxidation-assisted chemical recession, respectively. Fig. 6b shows that all the predicted total TCF strains at the different stress levels have alike three-stage characteristics. These TCF strain curves start from the almost same initial thermal expansion strain, and then are separated by the different strain rate, the different strain to saturation and the different time to failure. As we know, when a constant load is applied to C/SiC at elevated oxidizing temperatures, the applied stress opens the as-fabricated cracks and allows for easier ingress of oxygen to the fibers. At sufficiently high stress, cracks may be open too wide for crack closure and sealing to occur. Thus the applied stress will in turn play a critical role on the oxidation process of the internal fibers by forming the wider crack opening displacement  $\delta$ . In these TCF strain curves, the effects of stresses on the lives of the composites can be clearly seen. As expected, the predicted TCF strains at higher stresses fall to the left side of the plot while those at lower stresses are concentrated to the right side of the plot. Another effect from stress is the strain to saturation. The predicted TCF

strains at the higher stresses exhibit the pronounced higher strains to saturation in the secondary regime.

#### 4.4. Comparison with experimental results

Using the model of Eq. (20) and data listed in the Table 2, the calculational and experimental results of the TCF strain for the tested C/SiC composite specimen during 70 thermal cycles with  $\Delta T = 300^\circ\text{C}$  and fatigue stress of  $60 \pm 20$  MPa are presented in Fig. 7a. It can be seen that the model gives very accurate prediction. Furthermore, in the present investigation, the total strain of the material is depicted as the sum of a linear contribution corresponding to reversible deformation (i.e., thermal strain owing to heating/cooling and mechanical strain owing to reloading/unloading) and a non-linear contribution corresponding to the irreversible deformation resulting from the physical and chemical damage. Hence, the throughout strain response curve of the composite can be greatly simplified as the

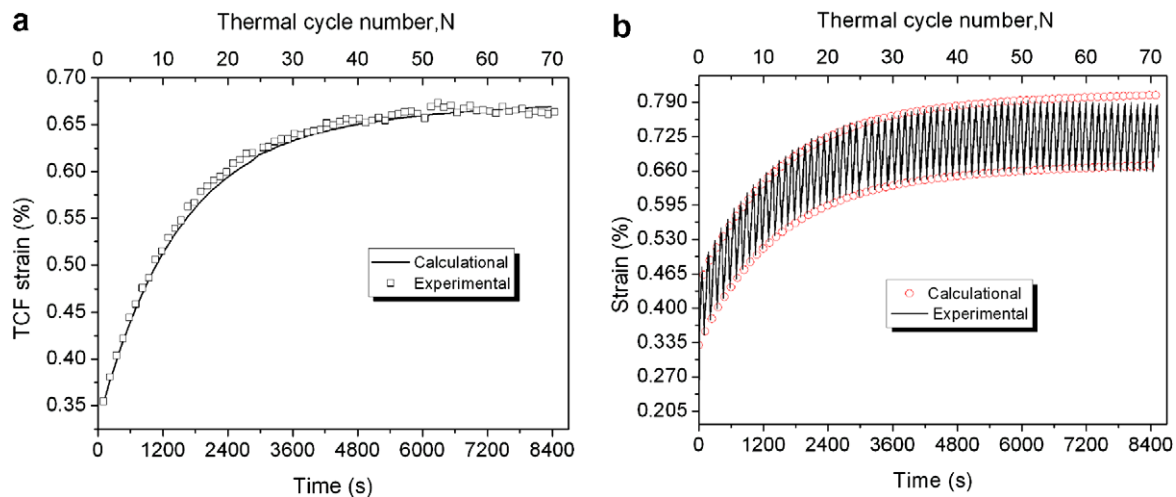


Fig. 7. Comparison of calculational results to experimental observations for the tested C/SiC composite specimen during 70 thermal cycles with  $\Delta T = 300^\circ\text{C}$  and fatigue stress of  $60 \pm 20$  MPa; (a) TCF strain and (b) entire strain response curve.

sum of the thermal strain, mechanical strain and TCF strain, related to test time  $t$ , as

$$\varepsilon^* = \varepsilon_{\text{TCF}}^T(n\theta) + \frac{E_m V_m \alpha_m + E_f V_f \alpha_f}{E_m V_m + E_f V_f} \dot{T}(t - n\theta) \pm \frac{\Delta\sigma_A}{2(E_m V_m + E_f V_f)} \quad (21)$$

where  $n$  is thermal cycle number (i.e., 0, 1, 2, 3, ...) and can be obtained from a integer conversion, with respect to the time  $t$  and period  $\theta$ , as

$$n = \left[ \frac{t}{\theta} \right] \quad (22)$$

Fig. 7b plots the experimental variation and predicted peak results of the total strain for the tested C/SiC composite specimen as a function of test time. Despite slight over prediction at the end of the curve, it is clear, without any attempt at fitting, that the strain model of Eq. (21) not only matches the order of the observed strain, but also follows the correct trend with increasing strain. The slight deviation between the model and experimental data is likely to be ascribed to the fact that the significant CTE parameter in this model also undergoes degradation with damage aggravation of the composites cycle by cycle. In the interest of simplicity but without losing generality, in contrast to the experimental observations in Fig. 1, the entire strain prediction illustrated in Fig. 8 exhibits a intrinsic nature of the thermal cycling strain response of the C/SiC composite under mechanical stress and oxidizing atmosphere, although neglecting the actual thermal inertia and linearly treating thermal strain and mechanical strain.

## 5. Summary and conclusion

A strain response model has been developed for carbonaceous fiber-reinforced ceramic matrix composites based

on environmental parameters, material parameters as well as on micromechanical properties of fibres, matrix and interface. The model takes into account thermophysical and chemical effects on the strain evolution of the composite with time, and its formulation relies on the concept of contribution of the matrix cracking, multiplication, of the fiber debonding, sliding and oxidation, both individuals and combinations. The model was utilized in analysing the effect of the environmental parameters such as stress, cooling rate, oxidant partial pressure, etc. on composite performance, as well as in linking the micromechanical interactions with the macroscopic response of the material to applied deformation. The model, without any attempt at fitting, was successful in assessing the experimentally obtained thermal cycling strain response behaviour for a specific case of a C/SiC composite in presence of mechanical fatigue and oxidizing atmosphere.

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## Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.carbon.2007.06.051](https://doi.org/10.1016/j.carbon.2007.06.051).

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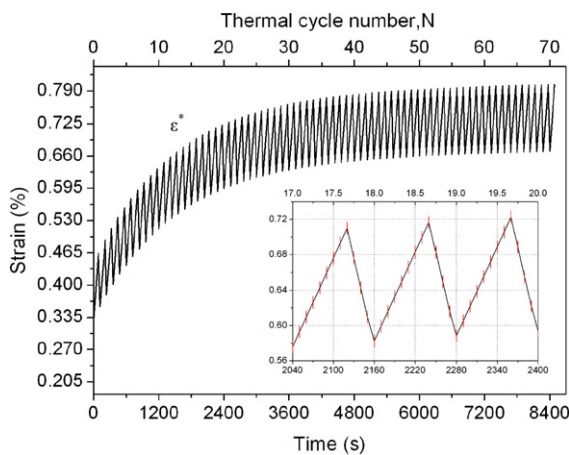


Fig. 8. Predicted entire strain response curve and a close-up of several representative thermal cycles, using the comprehensive strain model that is depicted as the sum of the TCF strain baseline, thermal strain and fatigue strain, when the composite is subjected to 70 thermal cycles with  $\Delta T = 300$  °C and fatigue stress of  $60 \pm 20$  MPa.

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